

implementing Activatⁿ funⁿ in code

$$2) \text{ Sigmoid} = \frac{1}{1 + e^{-z}} = \frac{1}{1 + \text{math.exp}(-z)}$$

case I:-

$$z = \text{value}(2.0)$$

$$= \frac{1}{1 + (\text{value}(2.73) ** (-z.data))}$$

→ 1st
design
choice

implementing from prev
ops

case II $z = \text{value}(2.0)$

$$z = \frac{1}{1 + \text{math.exp}(-z.data)}$$

$\text{Value}(z, (z,), 'sigmoid')$ → 2nd
design
choice

we are going with 2nd design choice

as it seems to be better choice bcz on backward
we had to do less # of ops as complex funⁿ

derivatives can be simple

if we go with design choice 1, then we will end up doing more # of ops, in a value type for $\times, +, -$ etc.

$$\tanh = \frac{e^z (e^z - e^{-z})}{e^z (e^z + e^{-z})} \quad \left(\begin{array}{l} \text{dividing entire} \\ e^z \text{ by } e^{-z} \\ \& \text{ multiply by } e^{-z} \end{array} \right)$$
$$= \frac{e^{2z} - 1}{e^{2z} + 1}$$

1st choice

$$I) \left(\frac{(\text{value}(2.71) \times \times (2 \times \text{self.data}) - 1)}{(\text{value}(2.71) \times \times (2 \times \text{self.data}) + 1)} \right)$$

Design I $\rightarrow$ Bad choice

as Back prop will be having so many int. calculations

$$z = \frac{\text{math.exp}(2 * \text{self.data}) - 1}{\text{math.exp}(2 * \text{self.data}) + 1}$$

Value(z, (self,), 'tan') → good design choice

1 on calculation of derivative of
 $\frac{d}{dz} \left(\frac{e^{2z} - 1}{e^{2z} + 1} \right) = (1 - z^2)$

→ so only
 2 optⁿ, (-),
 (*+)

So, design choice 2
 is better as total
 grad. calculation is less

→